Usage Guide

FSP Example Project Usage Guide

User's Manual

Renesas RX Family

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Renesas RX Family

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1. Introduction

This Example Project Usage Guide provides steps and guidelines for operating projects which use API and source generated by Renesas Flexible Software Package **(FSP)**.

1.1 Prerequisites

1. Tool experience: It is assumed that the user has prior experience working with integrated development environments, such as e^2 studio, and Tera Term.
2. Subject knowledge: It is assumed that the user has basic knowledge about microcontrollers, embedded systems, and FSP to modify the example projects. First time users are recommended to refer to FSP User Manual for Tutorial on *Starting Development*, paying special attention to sections *First RX MCU Project* – Blinky, *Importing an Existing Project into e2 studio,* and Tutorial on *Using HAL Drivers – Programming the WDT*.
3. Prior to running the example projects or programming the kits, all jumper settings must be set to the default settings. Refer to each kits' user's manual for the default jumper settings. After default jumper settings have been restored, refer the *<ep>_notes.md* for each example project to modify the values from the default.
4. The screen shots provided throughout this document are for reference. The actual screen content may differ depending on the version of software and development tools used.

2. Hardware and Software Requirements

RX FSP Example projects are designed to operate using RX MCU kits officially supported by Renesas. Supported kits are identified for each group of microcontrollers on the Rensas website.
Refer to the *<ep>_notes.md* file in the specific module folder of */example_projects* folder for additional hardware and software requirements for running the projects.

Note:

1. Some projects may require external hardware as mentioned in the respective *<ep>_notes.md* files
2. Some pin numbers may be printed on the back of the Evaluation Kit board.

Software Requirements

- Windows® 11 operating system

Below software requirements are provided as reference. Please refer to the Release Notes & Tools requirement for the FSP version of your choice to identify exact versions of software required.

- e^2 studio v2026-07 or later
- Renesas Flash Programmer V3.24.00 or later
- FSP v1.0.0 or later
- CC-RX Compiler: V3.08.00
- GCC for Renesas: 14.2.0.202607
- Renesas E2 and E2 Lite: 3.3.5
- SEGGER J-Link: 9.78
- Tera Term v4.99 or later

3. Tool Installation

3.1 FSP and tools installation

Refer to https://github.com/renesas/rx-fsp-examples/tree/master/example_projects/version_info_table.md to identify the version(s) of FSP needed to operate the example project(s) of interest. Download and install the latest version of FSP and tools from FSP GitHub repository.

1. Open FSP GitHub repository: https://github.com/renesas/rx-fsp
2. Go to the ***Releases*** section of Git and navigate to the desired FSP version.
3. Follow the instructions on installing and using FSP and e^2 studio.

3.2 Renesas Flash Programmer Installation

Download and install Tera Term Software for Windows from Renesas Flash Programmer (Programming GUI) | Renesas

1. Open the Renesas Flash Programmer download page.
2. Navigate to the Downloads section and download the desired Renesas Flash Programmer version.
3. Follow the instructions on installing and using Renesas Flash Programmer.

Defaults download path is `C:\Program Files (x86)\Renesas Electronics\Programming Tools`

Note: The supported debugger and communication interface depend on the selected RX board and device.

3.3 Tera Term Installation

Download and install Tera Term Software for Windows from Releases · TeraTermProject/teraterm

Default download path is `C:\Program Files (x86)\teraterm`.

4. Importing and Running the Project

4.1 Downloading the Project

4.1.1 Downloading the Project from GitHub

1. Open FSP Examples Repository: https://github.com/renesas/rx-fsp-examples/releases.
2. Navigate to **Assets** section.
3. Download the project bundle for specific kit. For e.g.: EK-RX74M-exampleprojects.pdf and ek_rx74m-exampleprojects.zip.

4.1.2 Downloading the Project from Renesas.com.

1. Download the example project bundle for the specific kit from https://www.renesas.com/us/en/products/software-tools/software-os-middleware-driver/software-package/rx-fsp.html#downloads

Note: Some example projects for a kit bundle may require an older version of FSP to be installed due to identified problems found in operating with latest version of FSP.

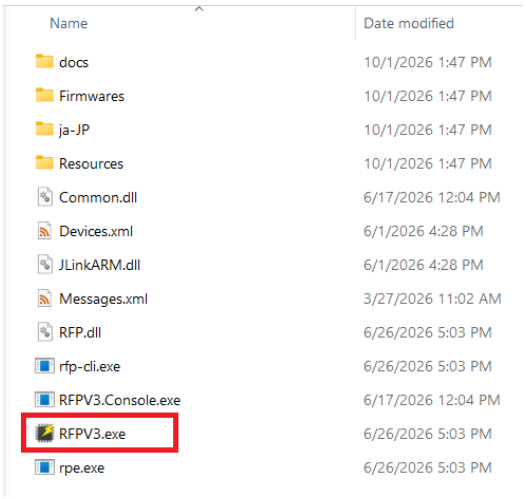
4.2 Running the project

There are two ways of running the project:

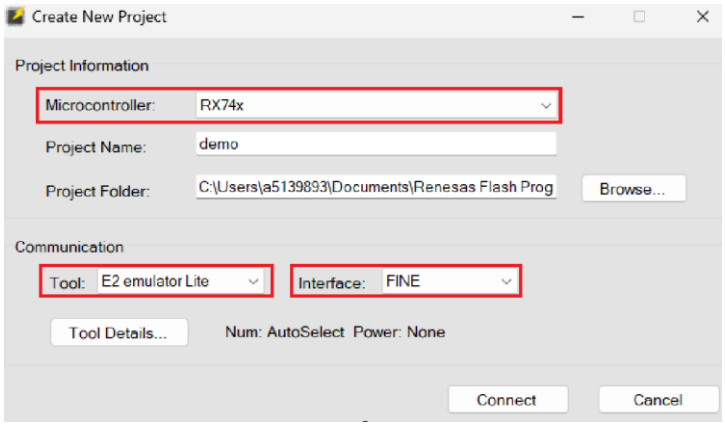
1. Flashing the pre-built binary (.hex file) and running the project is explained in section 4.2.1. This method is recommended when you are only interested in operating the microcontroller and evaluating the functionality described for the project.
2. Importing the project into e^2 studio, building, loading and running the project is explained in section 4.2.2. This method is recommended when you are interested in operating the microcontroller with a debugger attached to it and/or modifying the project to your needs.

4.2.1 Flashing pre-built binary

1. Navigate to the Renesas Flash Programmer installation folder: `C:\Program Files(x86)\Renesas Electronics\Programming Tools\Renesas Flash Programmer V3.xx` and open `RFPV3.exe`

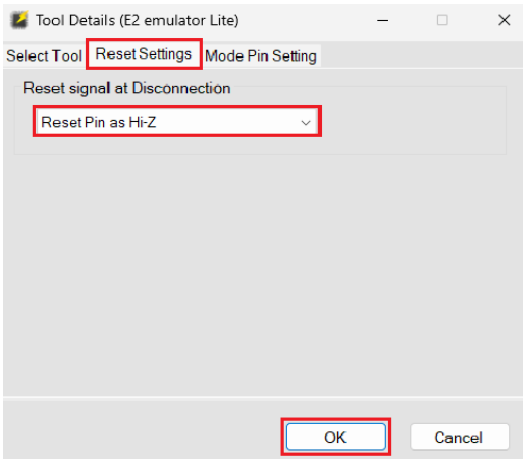


2. Select **File** > **New Project**. In the Communication Configuration dialog, configure the connection settings for the target device.



Note:

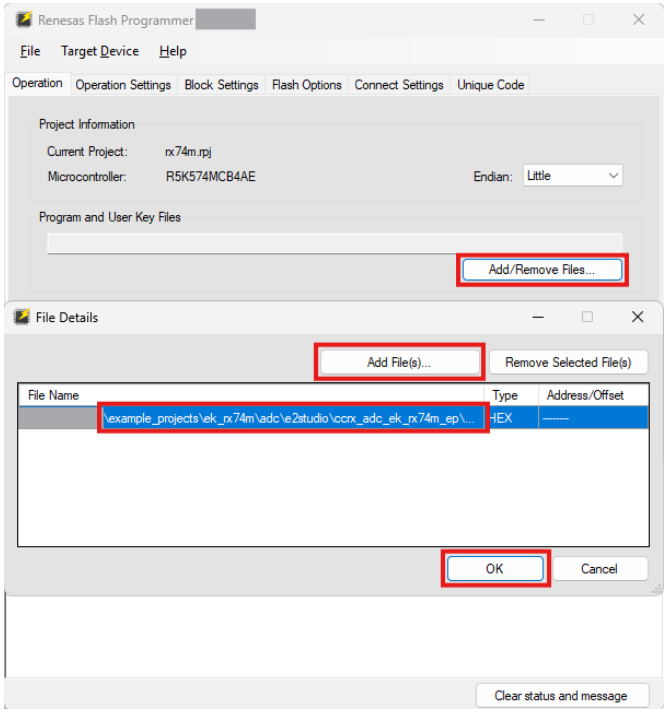
- Refer to the board user's manual or the <ep>_notes.md file for the correct communication settings.
- Example settings:
  - Microcontroller: RX74x
  - Tool: E2 emulator Lite
  - Interface: FINE



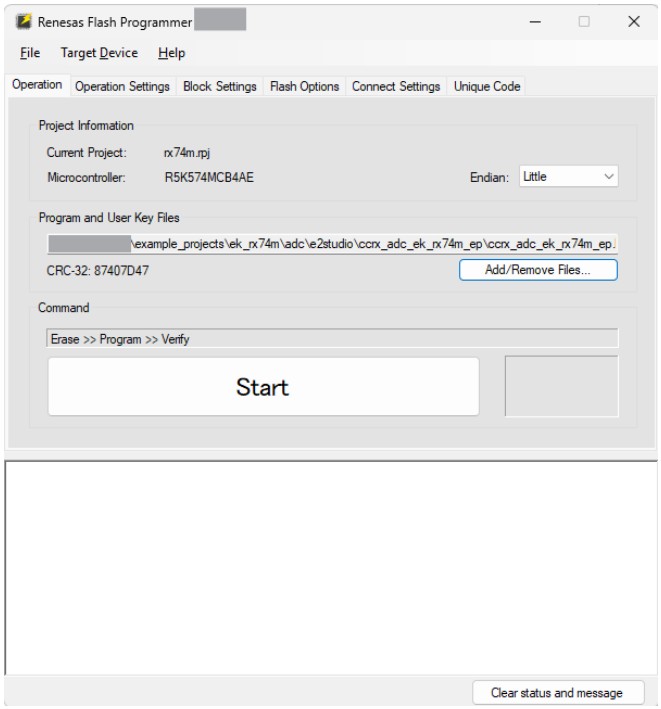
- If you want the target device to reset automatically after programming, click **Tool Details...** > **Reset Settings** and select **Reset Pin as Hi-Z** to reset board automatically after loading successfully.

Click **OK,** then click **Connect** to create the project.

3. Click **Add/Remove Files...** > **Add File(s)...**, browse to the location of the **.hex** file, and click **OK**. The selected file will be displayed in the Project Information window.



4. Click **Start**. When programming is complete, the status will be displayed as shown in the following figure.



5. Skip to Section Running the Project4.2.3 for steps on running the project.

4.2.2 Importing the project into e2studio

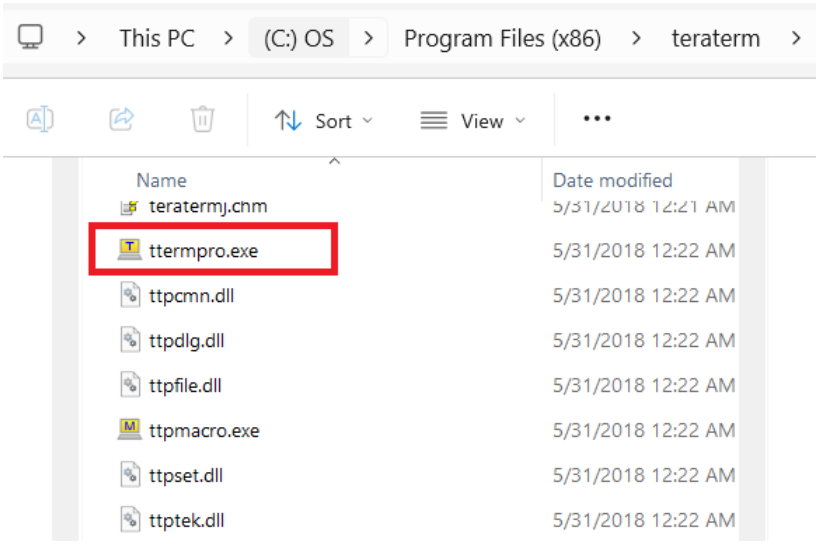
Refer to FSP User Manual for steps on importing a project:

- Importing an existing project
- Generating Project content
- Building the project
- Downloading the project image to the board

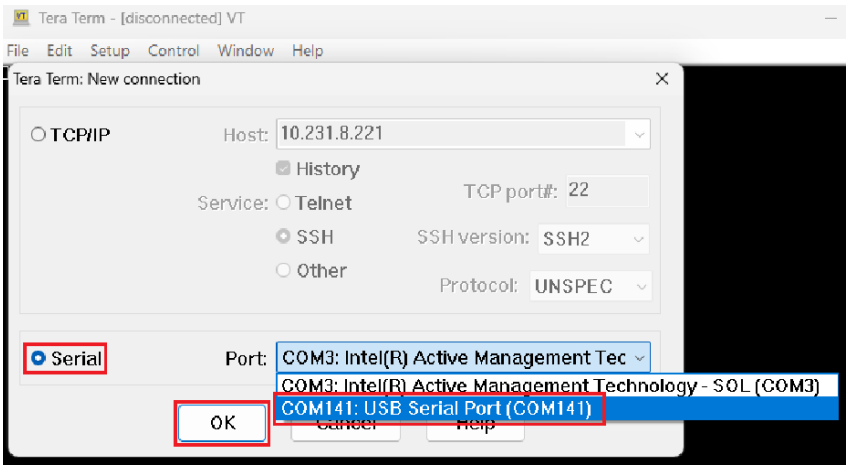
Then apply your learnings to the module/project of interest.

4.2.3 Running the Project

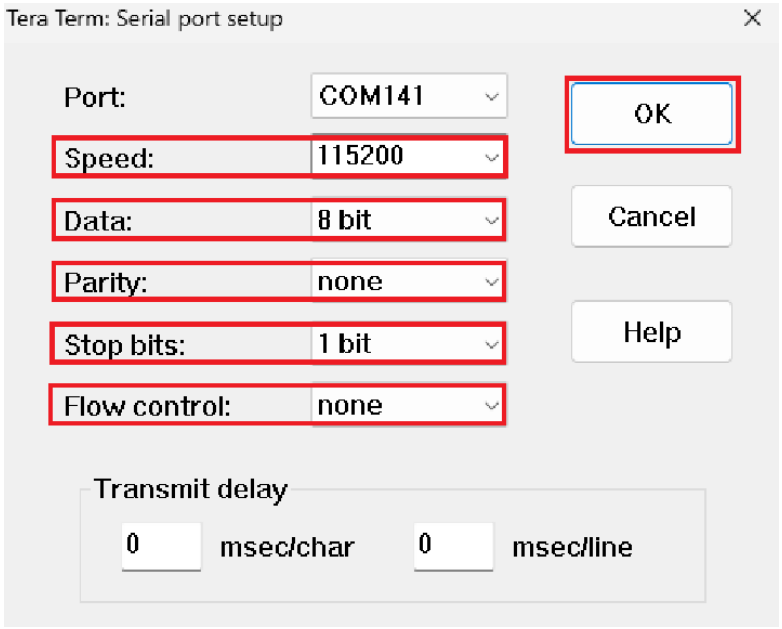
1. Open Tera Term by double-clicking `ttermpro.exe` in the Tera Term installation folder.



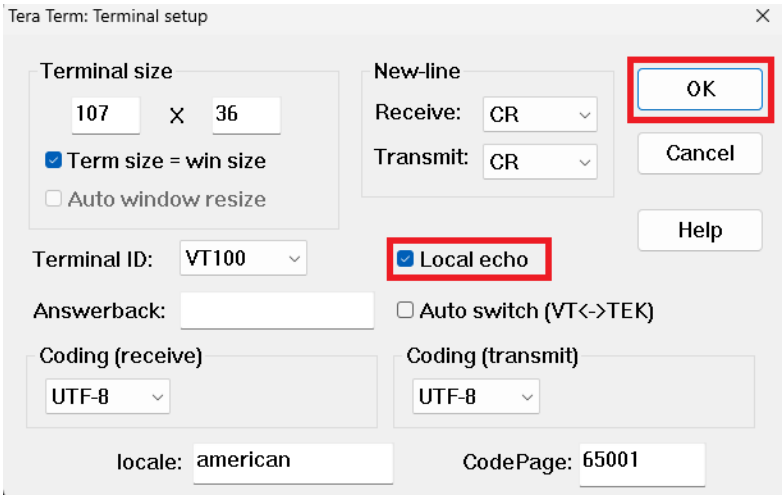
2. In the New connection dialog box, select **Serial**. From the Port drop-down list, select the **COM** port as-signed to the RX board kit, and then select **OK**.



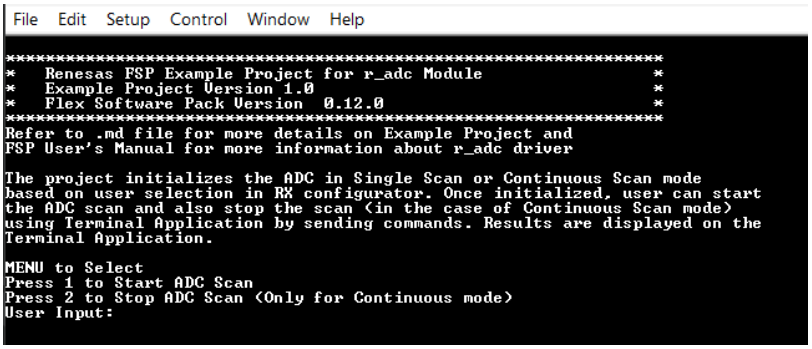
3. In **Tera Term**, select `Setup > Serial port…` Verify the serial port settings match the values below, and then select **OK**.
4. Configure the serial port settings as follow:
   - Baud rate: 115200 bps
   - Data length: 8 bits
   - Parity: None
   - Stop bit: 1 bit
   - Flow control: None



4. Select `Setup > Terminal…` and enable **Local echo** when typed input must be displayed.



5.To use the **RESET** push button, set **SW6-1** to **ON**, and then press the **RESET** push button to run the application. The application output is displayed in the **Tera Term** terminal.



5. References

FSP GitHub:	https://github.com/renesas/rx-fsp
FSP User Manual:	https://renesas.github.io/rx-fsp
	www.renesas.com/fsp#documents
FSP Example Projects:	https://github.com/renesas/rx-fsp-examples
Evaluation Kit Manuals:	
	http://www.renesas.com/rx/ek-rx74m
	http://www.renesas.com/rx/ek-rx74n
Support:	http://www.renesas.com/rx/support

Revision History

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		Page	Summary
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